SIMPLE ALGORITHM OF SLIDING AVERAGES WEB APP

A PREPRINT

Shahin Zibaee*
Unaffiliated

October 23, 2026

ABSTRACT

I describe a new web-based application for a simple algorithm, originally published in 2007. New additional features are incorporated to enable a user to better perform experiments using the algorithm. Some additional insights and discussion on the design and implementation of the algorithm itself are provided here and in a separate manuscript. Results of using the application for different protein properties are also presented in separate manuscripts.

Keywords sequence analysis · bioinformatics

1 Introduction

A Simple ALgorithm of Sliding Averages (SALSA) was originally designed for the purpose of estimating the propensity for cross-β-sheets found in amyloid fibrils for specific natively unfolded proteins. It uses a sliding window computation in which the property of interest is calculated for each window as an average over that window. Although it is a very straightforward sampling algorithm, its method includes unique aspects which, by their combination, appear to contribute insights into otherwise established characteristics of a protein, using only its amino acid sequence. Although originally designed for protein sequences, the web application places no restrictions on the type of sequence or property inputs SALSA will take, biological or otherwise.

2 Implementation details

The implementation was originally written in Java by the second author (O.S. Makin) of the original publication Zibaee et al. [2007]. I wrote a new implementation in Python (https://github.com/zibaee-2021/SALSA_mprops). The web application presented here was built with an agentic coding system in Python, Javascript, HTML and CSS. (The application is currently hosted on the free tier of Render). The code is not being made publicly accessible but the the application itself is free to use for academia.

2.1 Algorithm design

The following step-by-step description of SALSA's computation aligns with the step-by-step decisions made during its design and implementation:

1. A property of interest is calculated as a simple mean over a short contiguous stretch of the sequence.
2. This calculation is repeated on overlapping windows, sliding along the sequence one position at-a-time, akin to the long-established Kyte and Doolittle protein hydropathy algorithm Kyte and Doolittle [1982].
3. Unlike the Kyte and Doolittle method, the score of a window is then assigned to every position in the window, not just the central one.

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

4. The sliding window calculation is repeated from the start, using a different size of window. The typical use-case is to employ about 10 or more different window sizes.
5. The batch of scored windows represents the main output of SALSA. It was not displayed in the original publication, but is being made available in the new web application - displayed in 'Windows Plot'.
6. To aggregate the many scored windows into a more understandable and quantifiable summary format, the scores are summed as follows:
   (a) Lower-scoring windows are filtered out according to a user-defined threshold.
   (b) The sum of scores at each position is taken across every window covering that position. As such, for a sequence of length 100, the result is a list of 100 summed scores, each one corresponding to one specific position in the sequence.
   (c) The numbers can be plot against the sequence - displayed in 'Profile Plot'. This is second output.
7. A final summary score in the form of the sum of the summed scores is the third output. As such, this is a single number, regardless of sequence size.

My initial intention was that the scored windows output (as in 'Windows Plot') should be summarised in a way that does not lose information. However, I did not solve this seemingly impossible challenge. In the meantime the second author (O.S. Makin), who was also responsible for writing the original Java-based implementation of SALSA, came up with the summary computation (number 6 in the list above), which is displayed in Profile Plot. This turned out to be a very useful summary output and was crucial for the applying the algorithm in related studies.

$$\xi_{ij}(t) = P(x_t = i, x_{t+1} = j|y, v, w; \theta) = \frac{\alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})}{\sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})} \tag{1}$$

2.2 Additional functionality

2.2.1 Analysis parameters

2.2.2 Helical amphipathicity: Exploring a range of angles

to enable running experiments

Paragraph Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

3 Examples of citations, figures, tables, references

3.1 Citations

Citations use `natbib`. The documentation may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Here is an example usage of the two main commands (`citet` and `citep`): Some people thought a thing [Kour and Saabne, 2014a, Hadash et al., 2018] but other people thought something else [Kour and Saabne, 2014b]. Many people have speculated that if we knew exactly why Kour and Saabne [2014b] thought this...

3.2 Figures

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi. See Figure 1. Here is how you

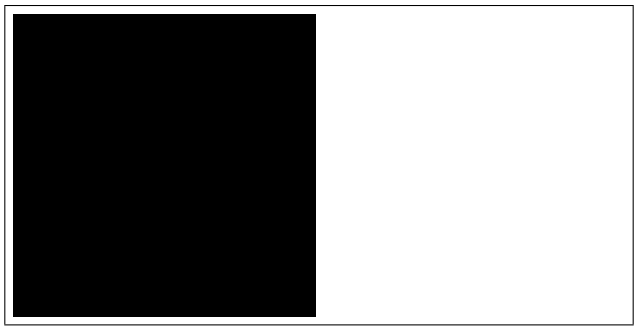


Figure 1: Sample figure caption.

Table 1: Sample table title

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

add footnotes. [2] Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetuer eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

3.3 Tables

See awesome Table 1.

The documentation for `booktabs` ('Publication quality tables in LaTeX') is available from:

`https://www.ctan.org/pkg/booktabs`

3.4 Lists

- Lorem ipsum dolor sit amet
- consectetur adipiscing elit.
- Aliquam dignissim blandit est, in dictum tortor gravida eget. In ac rutrum magna.

References

Shahin Zibaee, O. Sumner Makin, Michel Goedert, and Louise C. Serpell. A simple algorithm locates beta-strands in the amyloid fibril core of alpha-synuclein, abeta, and tau using the amino acid sequence alone. *Protein science : a publication of the Protein Society*, 16 5:906–18, 2007. URL `https://api.semanticscholar.org/CorpusID:9337360`.

Jack Kyte and Russell F. Doolittle. A simple method for displaying the hydropathic character of a protein. *Journal of molecular biology*, 157 1:105–32, 1982. URL `https://api.semanticscholar.org/CorpusID:6945799`.

George Kour and Raid Saabne. Real-time segmentation of on-line handwritten arabic script. In *Frontiers in Handwriting Recognition (ICFHR), 2014 14th International Conference on*, pages 417–422. IEEE, 2014a.

Guy Hadash, Einat Kermany, Boaz Carmeli, Ofer Lavi, George Kour, and Alon Jacovi. Estimate and replace: A novel approach to integrating deep neural networks with existing applications. *arXiv preprint arXiv:1804.09028*, 2018.

[2] Sample of the first footnote.

George Kour and Raid Saabne. Fast classification of handwritten on-line arabic characters. In *Soft Computing and Pattern Recognition (SoCPaR), 2014 6th International Conference of*, pages 312–318. IEEE, 2014b. doi:10.1109/SOCPAR.2014.7008025.